



Statistics and Parameters

You can use data for one of two purposes: to summarize or to draw conclusions.

Distinguishing Between Descriptive and Inferential Statistics

Descriptive statistics are used to numerically summarize or represent a set of data. Here are some examples:

- A student's grade point average is 3.27.
- The median home value in a city is \$259,000.
- The ducks in a zoo have a mean weight of 1.2 kg, with a standard deviation of 0.17 kg.

Inferential statistics are used to draw conclusions or make predictions. Notice in these examples that the purpose is to take the information gained from the sample and generalize it to the population it came from.

- A governor wants to know whether the residents in his state approve of his recent actions. In a poll, about 74% of those surveyed said they approved.
- Scientists caught 16 of the ducks in a lake, measured their lengths, and released them. They report, "Our findings indicate that the average length of the ducks in the lake is about 58 cm."

► **Think About It** In descriptive statistics, the numbers used to describe the data set are certain. In inferential statistics, the numbers used to describe the population are "best estimates."

Finding Parameters and Statistics

A **parameter** is a measurement that describes a population. A **statistic** is a measurement that describes a sample. Different notations make it easy to distinguish between parameters and statistics.

	Population	Sample
Mean	μ	$\bar{x}$
Standard deviation	σ	s
Proportion	p	$\hat{p}$

► Think About It

NOTATION The symbols μ and σ are the lowercase Greek letters mu and sigma. Read the symbol $\hat{p}$ as “ p -hat.”

At a zoo, 55% of the penguins are male. A researcher selects a sample of 5 of the penguins and notes that 3 are male. What are p and $\hat{p}$ for male penguins?

$$p = 0.55 \text{ because 55\% of the population is male}$$

$$\hat{p} = 0.6 \text{ because } \frac{3}{5}, \text{ or 60\%, of the sample is male}$$

The mean weight of the penguins at the zoo is 9.9 lb, with a standard deviation of 1.4 lb. The penguins in the researcher’s sample weigh 9.7 lb, 10.4 lb, 11.5 lb, 8.6 lb, and 11.8 lb. Find μ , $\bar{x}$, σ , and s .

Because they describe the population, $\mu = 9.9$ and $\sigma = 1.4$.

Calculate to find the statistics.

$$\bar{x} = \frac{52}{5} = 10.4 \text{ and } s = \sqrt{\frac{6.9}{4}} \approx 1.3$$

► **Remember** $s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$

Using a Table for a Simulation

In the zoo ponds, 20% of the goldfish are white. Use the Table of Random Digits in the appendix, starting at line 118, to simulate randomly selecting a sample of 40 goldfish and recording their color.

Because p is 20%, let 20% of the digits in the table represent selecting a white goldfish. For instance, let the digits 1 and 2 represent selecting a white goldfish and the digits 3, 4, 5, 6, 7, 8, 9, and 0 represent selecting a goldfish of another color.

116	84307	05017	79713	19252	39557	15608	90737	81430
117	43887	37323	47413	50487	23703	66932	09024	19395
118	01178	69415	09264	17728	58858	76330	57712	66411
119	36748	24247	75430	36452	95641	00723	22490	51418
120	66805	86573	49093	14317	20330	09825	36777	82572

The first five digits in line 118 are 0, 1, 1, 7, and 8, so the second and third fish selected were white. Continue reading across until you have selected 40 fish. You should have found that 10 of the 40 fish were white, so $\hat{p}$ is 25%.

► Q&A

Q Starting at line 118, what is $\hat{p}$ if you read by twos, letting the numbers 01 through 20 represent selecting a white goldfish?

A $\hat{p} = 27.5\%$ (11 out of 40)

Using a Table to Estimate a Probability

A dog successfully performs a trick 65% of the time. Use the Table of Random Digits in the appendix to estimate the probability that it will successfully perform the trick on at least 7 out of its next 10 attempts. Start at line 116 and perform 6 trials.

Let the numbers 01 through 65 represent a success. The first 10 two-digit numbers in line 116 are 84, 30, 70, 50, 17, 79, 71, 31, 92, and 52, representing 5 successes in 10 attempts.

Continuing along line 116, the next 10 numbers represent 6 successes in 10 attempts. Confirm for yourself that in the next 4 groups of 10 there are 6, 6, 7, and 7 successes. Out of the 6 trials, only 2 had at least 7 successes. Based on this simulation, the probability that the dog successfully performs the trick on at least 7 out of its next 10 attempts is about 33%.